

Aoyu Liu

B.S. in Mathematics and Computer Science, University of Electronic Science and Technology of China

Email: aoyuliu01@std.uestc.edu.cn [\[google scholar\]](#) Yingcai Honors College UESTC Image Processing Lab

PROFILE

By the time of my sophomore year, my research focused on low-level computer vision. I work under the guidance of [Prof. Shuaicheng Liu](#) in the Image Processing Lab at UESTC. During this period, I am fortunate to work closely with [Dr. Zhen Liu](#).

I also have practical experience in research computing operations and currently serve as the administrator of two 8-GPU RTX 4090 servers. My research interests are now expanding from low-level computer vision toward embodied intelligence, with a particular focus on manipulation and vision-language-action models.

GPA: 3.8/4.0. CET-4: 602. CET-6: 582.

Low-level Computer Vision

Embodied Intelligence

Manipulation

Vision-Language-Action Models

HDR Reconstruction

EXPERIENCE

- **2024.09 - Present:** Undergraduate Student, Yingcai Honors College, University of Electronic Science and Technology of China.

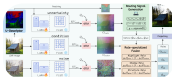
SELECTED COURSEWORK

- **Advanced Algebra and Geometry:** 85/100
- **Fundamentals of Programming:** 92/100
- **Computer Organization:** 93/100
- **Advanced Calculus:** 91/100
- **Introduction to Artificial Intelligence:** 86/100
- **Probability Theory and Mathematical Statistics:** 82/100

NEWS

- **2026:** Our work *Unleashing Pretrained Generative Priors for Single-Image HDR Reconstruction via Uncertainty-Routed Attention* is in submission to **SIGGRAPH Asia / ACM TOG**.
- **2026:** Our paper *ExpoCM: Exposure-Aware One-Step Generative Single-Image HDR Reconstruction* is accepted to **CVPR 2026**.
- **2025:** I received the University Special Scholarship from the University of Electronic Science and Technology of China.

SELECTED PUBLICATIONS

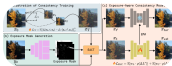


Unleashing Pretrained Generative Priors for Single-Image HDR Reconstruction via Uncertainty-Routed Attention

Aoyu Liu, Zhen Liu, Ziyi Wang, Bing Zeng, Shuaicheng Liu

ACM Transactions on Graphics (TOG), SIGGRAPH Asia, in submission

Research on pretrained generative priors for single-image HDR reconstruction with uncertainty-routed attention.



ExpoCM: Exposure-Aware One-Step Generative Single-Image HDR Reconstruction

Aoyu Liu, Zhen Liu, Ziyi Wang, Dian Chen, Bing Zeng, Shuaicheng Liu

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

Research on exposure-aware one-step generative reconstruction for single-image HDR imaging.

arXiv

COLLABORATORS

- Zhen Liu
- Ziyi Wang
- Shuaicheng Liu